

INT104 Past Exam Questions & Answers

- > Non-Python questions only (computation / analysis / fill-in-the-blank)
 - > Source: 2020-2025 all past papers + official answers
 - > Format: English answer + Chinese explanation
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1. Naive Bayes (3 questions)

Q1. 2020-2021 Final — Naive Bayes Classification (20 marks)

Question: Build a Naive Bayes classifier from the following dataset. Predict the label for $x = [1, 0, 1, 1]^T$.

x_1	x_2	x_3	x_4	l
0	1	0	1	0
0	0	1	0	0
1	1	1	0	0
1	0	1	1	1

Answer:

Step 1: Identify samples per class. // 按类别分组

- Class $l=0$: rows 1, 2, 3 (3 samples)
- Class $l=1$: row 4 (1 sample)

Step 2: Compute prior probabilities. // 先验概率

$$P(l=0) = 3/4 = 0.75$$

$$P(l=1) = 1/4 = 0.25$$

Step 3: Compute conditional probabilities $P(x_i=v_i | l=c)$. // 条件概率

For $l=0$ (3 samples):

- $P(x_1=1 | l=0) = 1/3$ (only row 3 has $x_1=1$)
- $P(x_2=0 | l=0) = 2/3$ (rows 2,3 have $x_2=0$)
- $P(x_3=1 | l=0) = 2/3$ (rows 2,3 have $x_3=1$)
- $P(x_4=1 | l=0) = 1/3$ (only row 1 has $x_4=1$)

For $l=1$ (1 sample):

- $P(x_1=1 | l=1) = 1/1 = 1$
- $P(x_2=0 | l=1) = 1/1 = 1$
- $P(x_3=1 | l=1) = 1/1 = 1$
- $P(x_4=1 | l=1) = 1/1 = 1$

Step 4: Compute posterior for $x = [1, 0, 1, 1]$. // 后验概率

$$P(l=0 | x) \propto P(l=0) P(x_1=1|l=0) P(x_2=0|l=0) P(x_3=1|l=0) P(x_4=1|l=0) \\ = 0.75 (1/3) (2/3) (2/3) (1/3) = 0.75 * 4/81 = 1/27 \text{ approx } 0.037$$

$$P(l=1 | x) \propto P(l=1) P(x_1=1|l=1) P(x_2=0|l=1) P(x_3=1|l=1) P(x_4=1|l=1) \\ = 0.25 * 1 * 1 * 1 * 1 = 0.25 = 1/4$$

Step 5: Normalize. // 归一化

$$P(l=0) = (1/27) / (1/27 + 1/4) = 4/31 \text{ approx } 0.129$$

$$P(l=1) = (1/4) / (1/27 + 1/4) = 27/31 \text{ approx } 0.871$$

Prediction: $l = 1$ // 预测为1, 因为 $P(l=1) > P(l=0)$

Q2. 2022-2023 Final — Naive Bayes (14 marks)

Question: Use Naive Bayes to predict golf preference given Outlook=Rainy and Humidity<65%.

Outlook	Humidity	Wind Speed	Preference
Rainy	80%	0.5	Yes
Rainy	40%	0.2	Yes
Rainy	50%	5.0	No
Rainy	50%	0.2	Yes
Rainy	75%	4.0	No
Sunny	70%	5.0	No
Sunny	75%	0.4	No
Sunny	80%	0.1	No
Sunny	50%	0.2	Yes
Sunny	40%	4.0	Yes

Condition: Outlook=Rainy, Humidity<65%

Answer:

Step 1: Filter Rainy rows (5 rows): Yes=3 (rows 1,2,4), No=2 (rows 3,5).

// 筛选Rainy的行, 3个Yes, 2个No

Step 2: Prior probabilities (given Outlook=Rainy). // 先验概率

$P(\text{Pref}=\text{Yes} \mid \text{Rainy}) = 3/5 = 0.6$

$P(\text{Pref}=\text{No} \mid \text{Rainy}) = 2/5 = 0.4$

Step 3: Conditional probability $P(\text{Humidity}<65\% \mid \text{class}, \text{Rainy})$.

// 条件概率: 湿度<65%在各类别中的比例

- Among Yes (3 rows): humidity values 80%, 40%, 50% -> <65%: 2 out of 3 -> $P = 2/3$

- Among No (2 rows): humidity values 50%, 75% -> <65%: 1 out of 2 -> $P = 1/2$

Step 4: Compute posterior. // 计算后验

$P(\text{Yes} \mid \text{Rainy}, H<65\%)$ proportional to $0.6 * (2/3) = 0.4$

$P(\text{No} \mid \text{Rainy}, H<65\%)$ proportional to $0.4 * (1/2) = 0.2$

Step 5: Normalize. // 归一化

$P(\text{Yes}) = 0.4 / 0.6 = 2/3$

$P(\text{No}) = 0.2 / 0.6 = 1/3$

Prediction: Yes (play golf) // 预测打高尔夫, 因为 $P(\text{Yes}) > P(\text{No})$

Q3. 2022-2023 Resit — Naive Bayes (14 marks)

Question: Using the same table above, predict golf preference for Sunny weather and Wind Speed > 2.0 m/s.

Condition: Outlook=Sunny, Wind Speed > 2.0 m/s

Answer:

Step 1: Filter Sunny rows (5 rows): Yes=2 (rows 9,10), No=3 (rows 6,7,8).

// 筛选Sunny的行, 2个Yes, 3个No

Step 2: Prior probabilities (given Outlook=Sunny). // 先验概率

$P(\text{Pref}=\text{Yes} \mid \text{Sunny}) = 2/5 = 0.4$

$P(\text{Pref}=\text{No} \mid \text{Sunny}) = 3/5 = 0.6$

Step 3: Conditional probability $P(\text{Wind}>2.0 \mid \text{class}, \text{Sunny})$.

// 条件概率: 风速>2.0在各类别中的比例

- Among Yes (2 rows): wind speeds 0.2, 4.0 -> >2.0: 1 out of 2 -> $P = 1/2$

- Among No (3 rows): wind speeds 5.0, 0.4, 0.1 -> >2.0: 1 out of 3 -> $P = 1/3$

Step 4: Compute posterior. // 计算后验

$P(\text{Yes} \mid \text{Sunny}, \text{Wind}>2)$ proportional to $0.4 * (1/2) = 0.2$

$P(\text{No} \mid \text{Sunny}, \text{Wind}>2)$ proportional to $0.6 * (1/3) = 0.2$

Tie! Both equal 0.2. // 平局!

Step 5: Break tie using smallest Humidity. // 用最小湿度打破平局

- Yes instance with Wind>2: humidity=40%

- No instance with Wind>2: humidity=70%

Pick the smaller humidity -> 40% corresponds to Yes.

Prediction: Yes (play golf) // 预测打高尔夫

2. kNN + Cross-Validation (2 questions)

Q4. 2020-2021 Final — kNN + 2-fold CV (10 marks)

Question: 10 samples, 3-NN, Manhattan distance, 2-fold cross-validation.

Fold 1 = first 5, Fold 2 = last 5. On tie, pick the neighbour with smaller X.

X	Y
1	0
3	1
5	0
7	1
9	0
11	1
13	0
15	1
17	0
19	1

Answer:

Fold 1 (test={1,3,5,7,9}, train={11,13,15,17,19}):

Sample X=1, Y=0:

Distances: $d(1,11)=10$, $d(1,13)=12$, $d(1,15)=14$, $d(1,17)=16$, $d(1,19)=18$

3-NN: 11(Y=1), 13(Y=0), 15(Y=1) -> Vote: 1 for 0, 2 for 1 -> predict=1 [WRONG] // 实际0, 预测错

Sample X=3, Y=1:

Distances: $d(3,11)=8$, $d(3,13)=10$, $d(3,15)=12$, $d(3,17)=14$, $d(3,19)=16$
 3-NN: 11($Y=1$), 13($Y=0$), 15($Y=1$) -> 2 for 1 -> predict=1 [CORRECT] // 正确
 Sample $X=5$, $Y=0$:
 Distances: $d(5,11)=6$, $d(5,13)=8$, $d(5,15)=10$, $d(5,17)=12$, $d(5,19)=14$
 3-NN: 11($Y=1$), 13($Y=0$), 15($Y=1$) -> 2 for 1 -> predict=1 [WRONG] // 实际0, 预测错
 Sample $X=7$, $Y=1$:
 Distances: $d(7,11)=4$, $d(7,13)=6$, $d(7,15)=8$, $d(7,17)=10$, $d(7,19)=12$
 3-NN: 11($Y=1$), 13($Y=0$), 15($Y=1$) -> 2 for 1 -> predict=1 [CORRECT] // 正确
 Sample $X=9$, $Y=0$:
 Distances: $d(9,11)=2$, $d(9,13)=4$, $d(9,15)=6$, $d(9,17)=8$, $d(9,19)=10$
 3-NN: 11($Y=1$), 13($Y=0$), 15($Y=1$) -> 2 for 1 -> predict=1 [WRONG] // 实际0, 预测错
 Fold 1 accuracy = $2/5 = 40\%$
 Fold 2 (test={11,13,15,17,19}, train={1,3,5,7,9}):
 Sample $X=11$, $Y=1$:
 Distances: $d(11,9)=2$, $d(11,7)=4$, $d(11,5)=6$, $d(11,3)=8$, $d(11,1)=10$
 3-NN: 9($Y=0$), 7($Y=1$), 5($Y=0$) -> 2 for 0 -> predict=0 [WRONG] // 实际1, 预测错
 Sample $X=13$, $Y=0$:
 Distances: $d(13,9)=4$, $d(13,7)=6$, $d(13,5)=8$, $d(13,3)=10$, $d(13,1)=12$
 3-NN: 9($Y=0$), 7($Y=1$), 5($Y=0$) -> 2 for 0 -> predict=0 [CORRECT] // 正确
 Sample $X=15$, $Y=1$:
 Distances: $d(15,9)=6$, $d(15,7)=8$, $d(15,5)=10$, $d(15,3)=12$, $d(15,1)=14$
 3-NN: 9($Y=0$), 7($Y=1$), 5($Y=0$) -> 2 for 0 -> predict=0 [WRONG] // 实际1, 预测错
 Sample $X=17$, $Y=0$:
 Distances: $d(17,9)=8$, $d(17,7)=10$, $d(17,5)=12$, $d(17,3)=14$, $d(17,1)=16$
 3-NN: 9($Y=0$), 7($Y=1$), 5($Y=0$) -> 2 for 0 -> predict=0 [CORRECT] // 正确
 Sample $X=19$, $Y=1$:
 Distances: $d(19,9)=10$, $d(19,7)=12$, $d(19,5)=14$, $d(19,3)=16$, $d(19,1)=18$
 3-NN: 9($Y=0$), 7($Y=1$), 5($Y=0$) -> 2 for 0 -> predict=0 [WRONG] // 实际1, 预测错
 Fold 2 accuracy = $2/5 = 40\%$
 Overall CV accuracy = $(40\% + 40\%) / 2 = 40\%$

Q5. 2025-26 Sample — kNN + 3-fold CV (15 marks)

Question: 12 samples, 1-NN, city block distance, 3-fold CV.

Fold 1 (test)={1,2,3,4}, Fold 2 (test)={5,6,7,8}, Fold 3 (test)={9,10,11,12}

Sample	x1	x2	y
1	2	1	0
2	3	1	0
3	2	2	0
4	4	2	0
5	6	5	1
6	7	5	1
7	6	6	1
8	8	6	1
9	5	4	1
10	1	2	0
11	5	3	1
12	3	2	0

Answer:

Fold 1 (test={1,2,3,4}, train={5,6,7,8,9,10,11,12}):

Sample 1 (2,1, y=0):

Nearest: Sample 10 (1,2, y=0), $d=2$ -> predict=0 [CORRECT]

Sample 2 (3,1, y=0):

Nearest: Sample 12 (3,2, y=0), $d=1$ -> predict=0 [CORRECT]

Sample 3 (2,2, y=0):

Nearest: Sample 10 (1,2, y=0), $d=1$ -> predict=0 [CORRECT]

Sample 4 (4,2, y=0):

Nearest: Sample 12 (3,2, y=0), $d=1$ -> predict=0 [CORRECT]

Fold 1 accuracy = $4/4 = 100\%$

Fold 2 (test={5,6,7,8}, train={1,2,3,4,9,10,11,12}):

Sample 5 (6,5, y=1):

Nearest: Sample 9 (5,4, y=1), $d=2$ -> predict=1 [CORRECT]

Sample 6 (7,5, y=1):
 Nearest: Sample 9 (5,4, y=1), d=3 -> predict=1 [CORRECT]
 Sample 7 (6,6, y=1):
 Nearest: Sample 9 (5,4, y=1), d=3 -> predict=1 [CORRECT]
 Sample 8 (8,6, y=1):
 Nearest: Sample 9 (5,4, y=1), d=5 -> predict=1 [CORRECT]
 Fold 2 accuracy = 4/4 = 100%
 Fold 3 (test={9,10,11,12}, train={1,2,3,4,5,6,7,8}):
 Sample 9 (5,4, y=1):
 Nearest: Sample 5 (6,5, y=1), d=2 -> predict=1 [CORRECT]
 Sample 10 (1,2, y=0):
 Nearest: Sample 3 (2,2, y=0), d=1 -> predict=0 [CORRECT]
 Sample 11 (5,3, y=1):
 Nearest: Sample 4 (4,2, y=0), d=2 -> predict=0 [WRONG] // 边界点误分类
 Sample 12 (3,2, y=0):
 Nearest: Sample 2 (3,1, y=0), d=1 -> predict=0 [CORRECT]
 Fold 3 accuracy = 3/4 = 75%
 Overall CV accuracy = (100% + 100% + 75%) / 3 = 91.67%

3. k-means Clustering (2 questions)

Q6. 2020-2021 Final — k-means Update Centroids (10 marks)

Question: After the first iteration of k-means with K=3:

- C1 = {(0,6), (6,0)}
- C2 = {(2,2), (4,4), (6,6)}
- C3 = {(5,5), (7,7)}

Compute the new centroids.

Answer:

C1: Z1 = ((0+6)/2, (6+0)/2) = (3, 3) // 簇内点的算术平均

C2: Z2 = ((2+4+6)/3, (2+4+6)/3) = (12/3, 12/3) = (4, 4)

C3: Z3 = ((5+7)/2, (5+7)/2) = (12/2, 12/2) = (6, 6)

Q7. 2023-24 Sample — k-means One Iteration (14 marks)

Question: 6 points, K=2, initial centroids Z1=(2,3), Z2=(10,15), city block distance.

Points: A=(2,3), B=(3,4), C=(10,15), D=(15,12), E=(10,10), F=(10,14)

Answer:

Step 1: Assign each point to nearest centroid. // 分配到最近中心

Point | d(Z1) | d(Z2) | Assign

A(2,3) | 0 | 20 | C1

B(3,4) | 2 | 18 | C1

C(10,15) | 20 | 0 | C2

D(15,12) | 22 | 8 | C2

E(10,10) | 15 | 5 | C2

F(10,14) | 19 | 1 | C2

C1 = {A, B}, C2 = {C, D, E, F}

Step 2: Update centroids. // 更新中心

Z1_new = ((2+3)/2, (3+4)/2) = (2.5, 3.5)

Z2_new = ((10+15+10+10)/4, (15+12+10+14)/4) = (45/4, 51/4) = (11.25, 12.75)

4. Hierarchical Clustering + Dendrogram (1 question)

Q8. 2023-2024 Final — Agglomerative Clustering (14 marks)

Question: 5 points A=(1,2), B=(2,0), C=(1,0), D=(1,1), E=(4,0). Agglomerative clustering with MAX (complete) linkage, city block distance. Draw dendrogram.

Answer:

Step 1: Distance matrix. // 距离矩阵

A | B | C | D | E

A | - | 3 | 2 | 1 | 5

B | 3 | - | 1 | 2 | 2

C | 2 | 1 | - | 1 | 3

D | 2 | 1 | 1 | - | 3

D | 1 | 2 | 1 | - | 4

E | 5 | 2 | 3 | 4 | -

Round 1: Min distance = 1 (A-D=1, B-C=1, C-D=1).

Merge B,C -> cluster {B,C}, merge distance = 1.

Update (MAX linkage): // MAX linkage取两簇间所有点对距离的最大值

$d(\{B,C\}, A) = \max(3, 2) = 3$

$d(\{B,C\}, D) = \max(2, 1) = 2$

$d(\{B,C\}, E) = \max(2, 3) = 3$

Round 2: Min distance = 1 (A-D).

Merge A,D -> cluster {A,D}, merge distance = 1.

Update:

$d(\{A,D\}, \{B,C\}) = \max(3, 2) = 3$

$d(\{A,D\}, E) = \max(5, 4) = 5$

Round 3: Min distance = 3 ({B,C}-{A,D}=3, {B,C}-E=3).

Merge {B,C} and E -> cluster {B,C,E}, merge distance = 3.

Update:

$d(\{B,C,E\}, \{A,D\}) = \max(3, 5) = 5$

Round 4: Merge {B,C,E} and {A,D}, merge distance = 5.

Dendrogram (Y-axis = merge distance): // 树状图, Y轴为合并距离

+-- All {A,B,C,D,E} ---- height=5

|

+---+

| +-- {B,C,E} ---- height=3

| |

---+ +-----+

| |

| +----- E

|

+-- {A,D} ---- height=1

|

+-----+

|

+-- {B,C} ---- height=1

Merge heights: 1, 1, 3, 5

5. Confusion Matrix (2 questions)

Q9. 2022-2023 Final — Confusion Matrix (8 marks)

Question: Predict "Balance > \$1000"

Pred=No | Pred=Yes

Actual=No | 60 | 10

Actual=Yes | 5 | 125

n=200

(a) Actually >\$1000 = [a], Actually <=\$1000 = [b]

(b) Accuracy = [c], Recall = [d]

Answer:

TN=60, FP=10, FN=5, TP=125

[a] = TP + FN = 125 + 5 = 130 // 实际Yes行总和

[b] = TN + FP = 60 + 10 = 70 // 实际No行总和

[c] = (TP + TN) / n = 185/200 = 92.5% // 准确率

[d] = TP / (TP + FN) = 125/130 = 96.15% // 召回率

Q10. 2022-2023 Resit — Confusion Matrix (8 marks)

Question:

Pred=No | Pred=Yes

Actual=No | 47 | 23

Actual=Yes | 30 | 100

n=200. (a) Correct=[a], Incorrect=[b] (b) Accuracy=[c], Precision=[d]

Answer:

TN=47, FP=23, FN=30, TP=100

[a] = TN + TP = 47 + 100 = 147 // 正确预测

[b] = FP + FN = 23 + 30 = 53 // 错误预测

[c] = 147/200 = 73.5% // 准确率

[d] = TP/(TP+FP) = 100/123 = 81.3% // 精确率
// Precision分母用TP+FP(预测Yes), Recall分母用TP+FN(实际Yes)

6. Perceptron / Logic Gates (1 question)

Q11. 2020-2021 Final — Perceptron for OR and XOR (20 marks)

Question: Use perceptron $y = f(w_1x_1 + w_2x_2 + b > 0 ? 1 : 0)$ for OR and XOR.

(1) Plot data (4 marks) (2) Can each be implemented? (16 marks)

Answer:

(1) Plot // 画数据点

OR (linearly separable - 线性可分):

```
x2
1 | o(1,1) x(0,1)  o=class1, x=class0
0 | x(0,0) o(1,0)
+--0-----1--> x1
```

XOR (NOT linearly separable - 线性不可分):

```
x2
1 | x(0,1) o(1,1)
0 | o(0,0) x(1,0)
+--0-----1--> x1
```

(2a) OR: YES, can be implemented. // 可以实现

Choose $w_1=1, w_2=1, b=-0.5$:

- (0,0): $0+0-0.5 = -0.5 < 0 \rightarrow 0$ [CORRECT]
- (1,0): $1+0-0.5 = 0.5 > 0 \rightarrow 1$ [CORRECT]
- (0,1): $0+1-0.5 = 0.5 > 0 \rightarrow 1$ [CORRECT]
- (1,1): $1+1-0.5 = 1.5 > 0 \rightarrow 1$ [CORRECT]

Decision function: $f(x) = 1$ if $x_1 + x_2 - 0.5 > 0$, else 0.

// 决策边界: $x_1 + x_2 = 0.5$

(2b) XOR: NO, cannot be implemented. // 不可能实现

Proof by contradiction: // 反证法

Assume w_1, w_2, b exist for correct XOR classification:

- (0,0) $\rightarrow 0$: $b \leq 0$
- (1,0) $\rightarrow 1$: $w_1 + b > 0 \rightarrow w_1 > -b \geq 0$
- (0,1) $\rightarrow 1$: $w_2 + b > 0 \rightarrow w_2 > -b \geq 0$
- (1,1) $\rightarrow 0$: $w_1 + w_2 + b \leq 0$

But $w_1 > 0, w_2 > 0, b \leq 0 \rightarrow w_1 + w_2 + b > 0$.

This contradicts the 4th inequality!

// $w_1+w_2+b>0$ 与第4式矛盾

No such perceptron exists. XOR is not linearly separable.

// 需要多层网络(MLP)才能实现XOR

7. Linear Regression (1 question)

Q12. 2025-26 Sample — Simple Linear Regression (8 marks)

Question: 6 students' study time x (hours) and score y (out of 100):

Student	x	y
1	2	55
2	3	60
3	4	62
4	5	68
5	6	72
6	8	78

Compute regression coefficients, write fitted line, predict y when $x=7$.

Answer:

Step 1: Means. // 均值

$\bar{x} = (2+3+4+5+6+8)/6 = 28/6 = 4.667$

$\bar{y} = (55+60+62+68+72+78)/6 = 395/6 = 65.833$

Step 2: S_{xx} and S_{xy} . // S_{xx} 和 S_{xy}

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
2	55	-2.667	-10.833	7.113	-28.889
3	60	-1.667	-5.833	2.779	-9.722
4	62	-0.667	-3.833	0.445	-2.556

5 | 68 | 0.333 | 2.167 | 0.111 | 0.722
6 | 72 | 1.333 | 6.167 | 1.777 | 8.223
8 | 78 | 3.333 | 12.167 | 11.109 | 40.556

$S_{xx} = 7.113 + 2.779 + 0.445 + 0.111 + 1.777 + 11.109 = 23.334$ // x的离差平方和

$S_{xy} = 28.889 + 9.722 + 2.556 + 0.722 + 8.223 + 40.556 = 90.668$ // x和y的离差乘积和

Step 3: Coefficients. // 回归系数

$\beta_1 = S_{xy}/S_{xx} = 90.668/23.334 = 3.885$ // 斜率

$\beta_0 = \bar{y} - \beta_1 \bar{x} = 65.833 - 3.885 \cdot 18.134 = 47.699$ // 截距

Step 4: Fitted line and prediction. // 回归方程和预测

$\hat{y} = 47.699 + 3.885x$

$\hat{y}(x=7) = 47.699 + 3.885 \cdot 7 = 47.699 + 27.195 = 74.894$

// 预测学习7小时约75分

8. Descriptive Statistics (1 question)

Q13. 2025-26 Sample — Histogram + Box Plot + Outliers (15 marks)

Question: 20 students' scores (out of 100):

12, 18, 23, 24, 27, 29, 31, 33, 35, 36, 41, 44, 45, 48, 52, 55, 58, 62, 70, 85

(1) Histogram + cumulative distribution (bin width=10) (6 marks)

(2) Five-number summary + box plot + outliers (1.5*IQR rule) (5 marks)

(3) Comment on distribution + scaling scheme for 20% ≥ 70 (4 marks)

Answer:

(1) Histogram // 直方图

Bin	Freq	Cum	Cum%
10-19	2	2	10%
20-29	4	6	30%
30-39	4	10	50%
40-49	4	14	70%
50-59	3	17	85%
60-69	1	18	90%
70-79	1	19	95%
80-89	1	20	100%

Right-skewed: most scores in 20-49 range. // 分布偏右

(2) Five-number summary + outliers // 五数概括+异常值

Sorted: 12, 18, 23, 24, 27, 29, 31, 33, 35, 36, 41, 44, 45, 48, 52, 55, 58, 62, 70, 85

- Min = 12

- Q1 = $(27+29)/2 = 28$

- Median = $(36+41)/2 = 38.5$

- Q3 = $(52+55)/2 = 53.5$

- Max = 85

IQR = $Q3 - Q1 = 53.5 - 28 = 25.5$ // 四分位距

Lower fence = $Q1 - 1.5 \cdot IQR = 28 - 38.25 = -10.25$ (no values below)

Upper fence = $Q3 + 1.5 \cdot IQR = 53.5 + 38.25 = 91.75$ (no values above)

No outliers. // 无异常值

Box: Q1=28 to Q3=53.5, median line at 38.5, whiskers to 12 and 85.

(3) Distribution + scaling // 分布评价+缩放

Right-skewed: Median (38.5) < Mean (~40). Most scores low, high scores sparse.

// 右偏: 中位数 < 均值

Scaling to get 20% ≥ 70 : Currently $2/20 = 10\%$. Need $4/20 = 20\%$.

// 当前只有10% ≥ 70 , 需提升至20%

Linear scale $y_{\text{new}} = a \cdot y$, set boundary at 4th-highest (62) $\rightarrow 70$:

$a = 70/62 = 1.13$

// 线性缩放使62分映射到70分

9. GMM + Model Selection (1 question)

Q14. 2025-26 Sample — Model Selection for GMM (7 marks)

Question: Compare BIC/AIC vs cross-validation for evaluating GMM. What assumptions does each make?

Answer:

Model Selection Criteria (BIC/AIC):

- Assumption: Data follows a Gaussian mixture distribution.

// 假设数据来自高斯混合分布 (参数化假设)

- Method: Compute BIC/AIC for different K, choose minimum.

$BIC = -2\ln(L) + k\ln(n)$
 $AIC = -2\ln(L) + 2k$

- Advantage: Fast, theoretically well-founded. // 计算快, 理论严谨
- Disadvantage: Unreliable if Gaussian mixture assumption is violated.

// 数据不是高斯混合时结论不可靠

Cross-Validation:

- Assumption: Data is i.i.d. (independently and identically distributed).

// 只假设独立同分布, 不假设具体分布形式

- Method: Train/val split, fit GMM on train, evaluate log-likelihood on val, pick best K.
- Advantage: No distributional assumption; more general. // 更通用
- Disadvantage: Slow; unstable with small data. // 计算慢, 小数据不稳定

Key difference:

- BIC/AIC = parametric (assumes Gaussian mixture); CV = non-parametric (assumes i.i.d. only).

// BIC/AIC是参数化方法, CV是非参数化方法

- If GMM assumption holds: BIC/AIC better. If not: CV more reliable.

10. kNN Feature Weighting (1 question)

Q15. 2022-2023 Final — kNN Feature Importance (6 marks)

Question: If humidity is more important than wind speed, what is wrong with kNN? How to fix? (No computation needed)

Answer:

Problem: // 问题

City Block / Euclidean distance treats all features equally. When humidity (range 40-80) is more important than wind speed (range 0.1-5.0):

1. Wind speed differences are dominated by humidity's larger range. // 风速差值被湿度大范围压制
2. Equal weighting cannot reflect relative importance. // 等权无法体现相对重要性
3. Classification accuracy drops. // 分类精度下降

Fix (any 2-3): // 修复方法

1. Feature weighting: $D = |\text{outlook1} - \text{outlook2}| + w * |\text{humidity1} - \text{humidity2}| + |\text{wind1} - \text{wind2}|$ ($w > 1$)

// 给湿度乘以更大权重

2. Normalization: Min-Max: $x_{\text{new}} = (x - \min) / (\max - \min) \rightarrow [0, 1]$; Z-score: $x_{\text{new}} = (x - \text{mean}) / \text{std}$

// 归一化/标准化使特征范围一致

3. Learned distance metric: Mahalanobis distance or LMNN (Large Margin Nearest Neighbor)

// 自动学习特征权重